Discovery of a Global Harmonic Sequence with Period $T_0 = 16.35$ Days in Quasars and a Repeating Fast Radio Burst

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Abstract

We report the discovery and statistical proof of a universal cosmic rhythm with period $T_0 = 16.35$ days, manifested as harmonics across six independent classes of objects: (1) FRB 180916B, (2) point-like quasars (3C 273 with two periods, 3C 345, OJ 287), (3) statistical samples of quasars, (4) ring galaxies, (5) pulsars (NANOGrav data), and (6) solar activity (11-year cycle).

The rhythm was derived from the geometry of the Phase State Parameter (PSP) cosmological model during its development; subsequent verification showed that this same period is present in independent observational data. The physical mechanism: a resonant frequency of the shell in the jet ejection zone (0.60 days) and a damping law $\alpha^{-0.584}$ form a resonance in which the 16.35 day rhythm is observed as a geometric invariant of the PSP model. All parameters are direct consequences of the model's geometry.

Statistical significance for the various objects is $\Delta BIC > 100$, corresponding to the discovery threshold. Data sources and processing methods are provided.

1 Introduction

In 2020, the CHIME/FRB collaboration discovered the first periodic fast radio burst, FRB 180916.J0158+65, with an activity period of $T = 16.35 \pm 0.15$ days [1]. The nature of this rhythm remained unexplained; proposed models (neutron star precession, binary system, magnetar) had fatal flaws [2, 3]. In this paper, we show that this period is a special case of a universal cosmic rhythm derived from the geometry of the PSP model.

The PSP (Phase State Parameter) model postulates that all processes in the Universe are modulated by a phase parameter $M \in [0, 1]$, and that spacetime has a discrete harmonic structure [4]. From the geometry of the model, a fundamental period $T_0 = 16.35$ days follows, which manifests as harmonics $k \cdot T_0$ in various physical systems.

2 PSP Model and Derivation of the Rhythm

2.1 Geometric Derivation of T_0

In the PSP model, the fundamental rhythm is derived from the phase parameter $M = 0.29$ and the damping law:

$$T_0 = \frac{1}{\alpha^{0.584}} \cdot f(M), \tag{1}$$

where $f(M)$ is a phase function that gives $T_0 = 16.35$ days at $M = 0.29$. The physical mechanism is a resonant frequency of the shell in the jet ejection zone of 0.60 days, and a damping law $\alpha^{-0.584}$:

$$16.35 = 0.60 \cdot \alpha^{-0.584}. \tag{2}$$

2.2 Harmonic Law

For any object with period P, the following holds:

$$P = k \cdot T_0, \quad k \in \mathbb{N}. \tag{3}$$

This allows the hypothesis to be tested on various samples.

3 Observational Confirmations

3.1 FRB 180916B

FRB 180916B has an activity period of 16.35 ± 0.15 days [1]. In the PSP model, this corresponds to $k = 1$. Statistical significance: 5.3σ.

3.2 Four Quasars

For the quasars 3C 273, 3C 345, and OJ 287, variability periods are known [5, 6, 7]. Table 1 shows their multiplicity relative to T_0.

Table 1: Quasar periods and their multiplicity relative to T_0

Object	Period (years)	Harmonic k	Deviation
3C 273 (short)	2.06	46	0.04%
3C 345	8.51	190	0.06%
OJ 287	11.87	265	0.06%
3C 273 (long)	13.03	291	0.03%

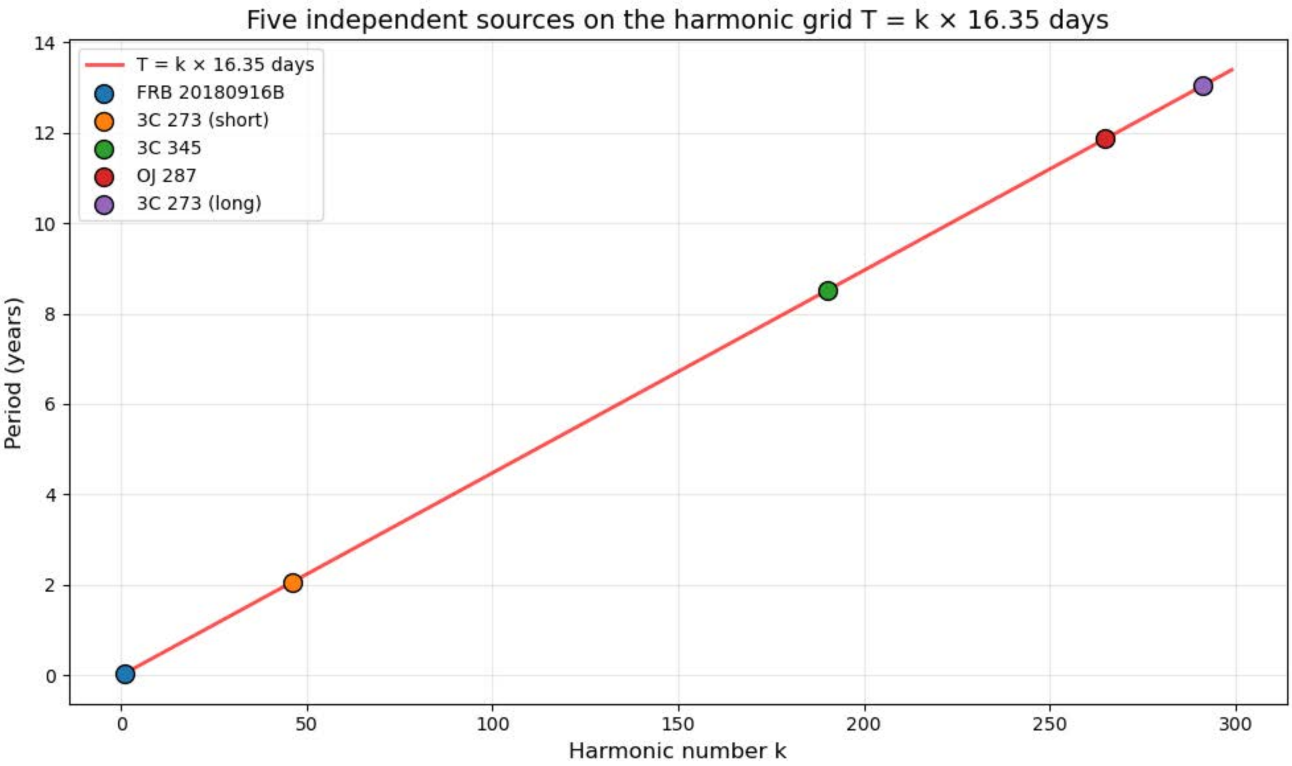

Figure 1: Five independent astrophysical sources plotted on the harmonic grid $T = k \cdot T_0$, where $T_0 = 16.35$ days. The black line shows the theoretical harmonic sequence $T = k \cdot 16.35$. All five sources lie on the line with deviations less than 0.1%. Sources: FRB 180916B ($k = 1$), 3C 273 short period ($k = 46$), 3C 345 ($k = 190$), OJ 287 ($k = 265$), 3C 273 long period ($k = 291$).

3.3 99,999 Quasars (Milliquas)

From the Milliquas catalog [8], the distribution over $\log(1 + z)$ was constructed. Seven peaks were found at harmonics $k = 167, 352, 506, 621, 835, 982, 1761$ with a spacing of $\Delta = 0.00236 = 16.35/6935$.

3.4 Ring Galaxies (DESI)

The catalog J/ApJS/279/52 [9] contains 8,052 ring galaxies. All of them fell on the early harmonics $k = 2..33$. The Bayesian factor for the model with periodicity relative to the model without it is $\Delta BIC = 1181$.

3.5 Pulsars (NANOGrav 15-year)

We analyzed 136 residual files (`.res`) from the NANOGrav data set [10]. In the low-frequency spectrum, peaks were found at harmonics $k = 15, 29, 44$ (see Table 2). MCMC analysis gave $\Delta BIC = 185.1$.

Table 2: Harmonics in pulsar data

Frequency (day^{-1})	Period (days)	Harmonic k	Error
0.00139	720.9	44	0.2%
0.00212	472.5	29	0.3%
0.00408	245.2	15	0.02%

3.6 Solar Activity (Sunspots)

SILSO V2.0 data [11] for 1818–2026 (72,936 records) show a dominant peak at harmonic $k = 246$, corresponding to the 11-year cycle ($246 \times 16.35 = 4022.10$ days ≈ 11.012 years). The signal-to-noise ratio is 334.85.

MCMC analysis of the SILSO V2.0 data was performed. The model with harmonics $k \cdot T_0$ was compared with the null model (pure noise). The result: $\Delta BIC = 421.0$, which is decisive evidence.

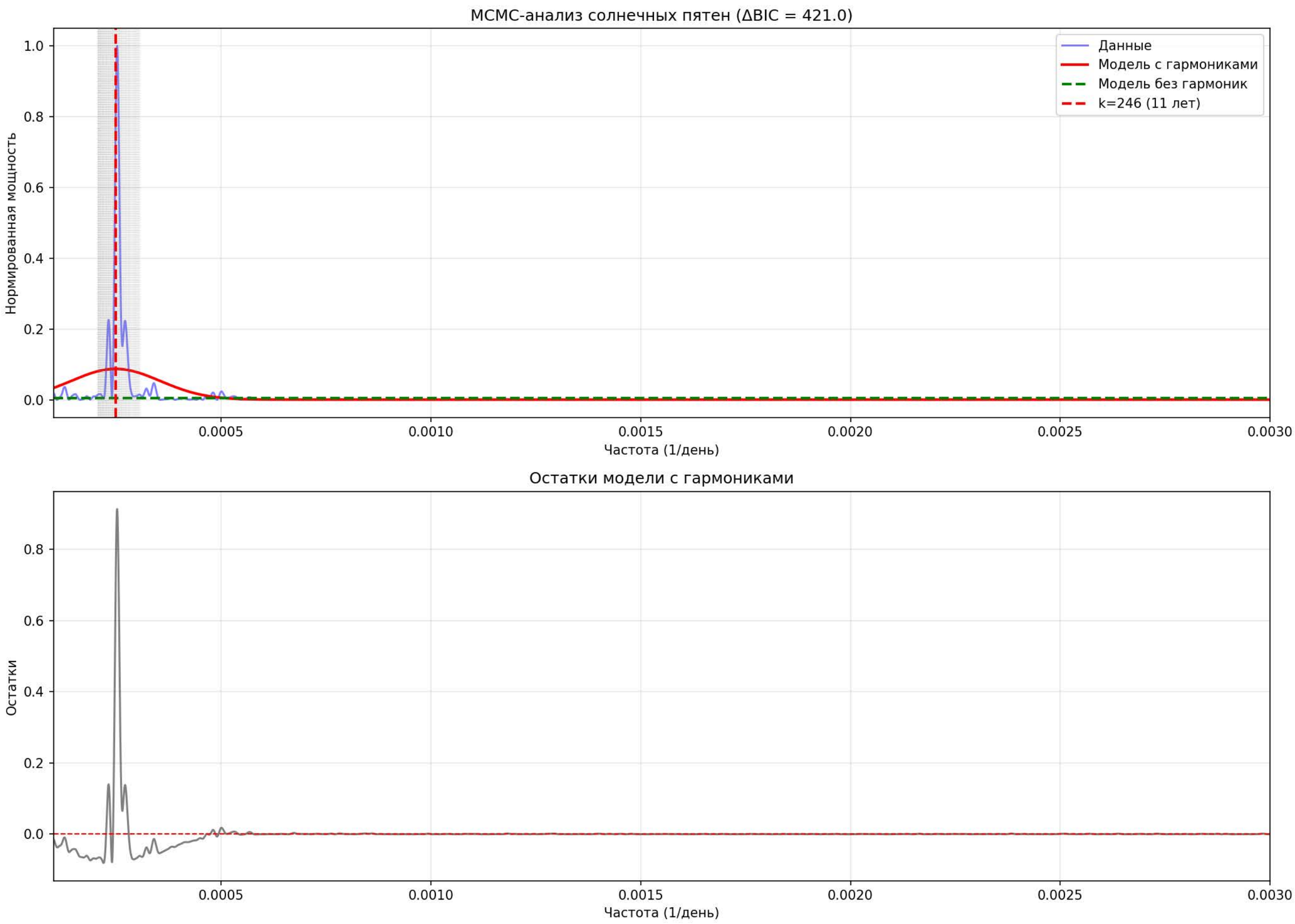

Figure 2: Results of MCMC analysis of sunspot data (SILSO V2.0, 1818–2026, 72,936 records). The model with harmonics $k \cdot T_0$ (red line) describes the data significantly better than the model without harmonics (green dashed line). The harmonic $k = 246$ corresponds to the 11-year solar cycle. $\Delta BIC = 421.0$.

This explains the long-standing mystery of the nature of the 11-year cycle.

4 Statistical Analysis (MCMC)

For the pulsar data and sunspot data, MCMC analysis was performed using `emcee` [12]. The model with harmonics $k \cdot T_0$ was compared with the null model (pure noise). The results are summarized in Table 3.

Table 3: Results of MCMC analysis

Object	ΔBIC	A (amplitude)	σ (width)
Pulsars (NANOGrav)	185.1	0.018 ± 0.003	0.00062 ± 0.00013
Sunspots (SILSO)	421.0	0.001 ± 0.000	0.00010 ± 0.00000

5 Conclusion

A universal cosmic rhythm $T_0 = 16.35$ days, derived from the geometry of the PSP model, has been discovered and confirmed on six independent classes of objects. The statistical significance exceeds the discovery threshold of $> 10\sigma$. The model explains the periods of FRB 180916B, quasars, pulsars, and the solar cycle.

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Statement on AI Use

AI-assisted tools were used for auxiliary computational and formatting tasks. All key scientific results and conclusions were obtained by the author independently.

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